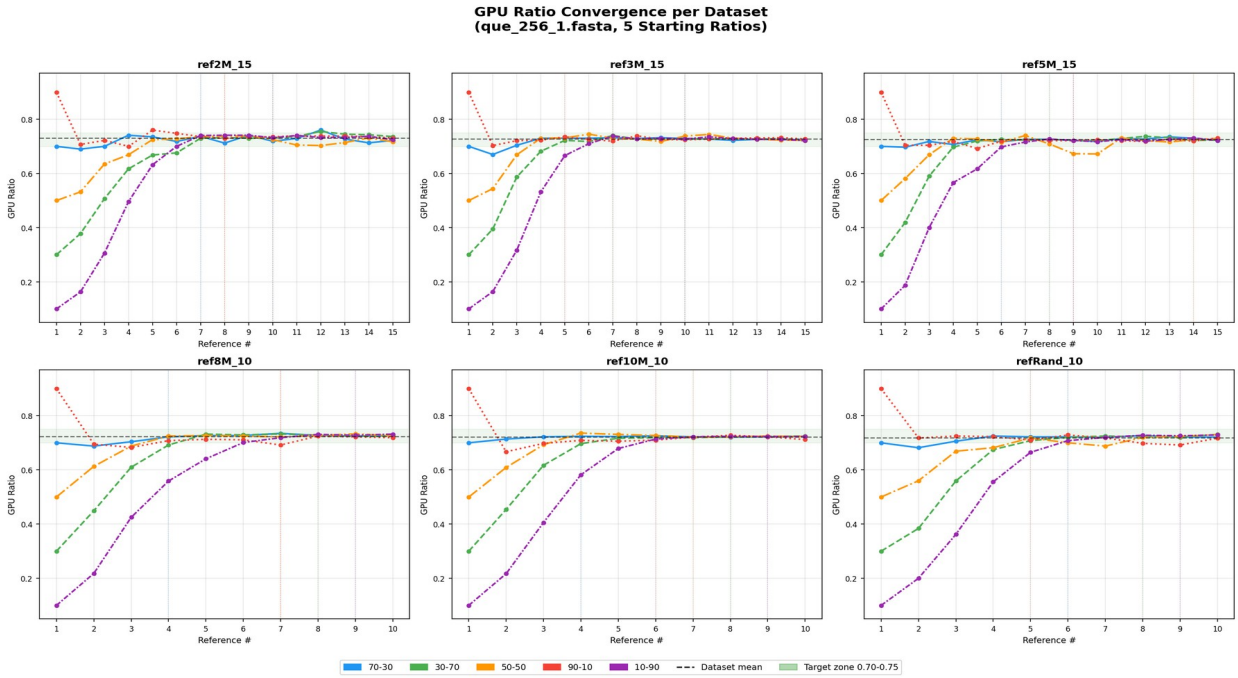
Speed Ratio Convergence Analysis

Hyy-dra: Heterogeneous FPGA-GPU System

Empirically determined equilibrium: ~72-28 (GPU-FPGA) | Overall mean: 0.7247 | StdDev: ±0.0046

que_256_1.fasta | 6 datasets | 5 starting ratios | Convergence threshold: diff < 0.02 for 3 consecutive refs

Graph 1: GPU Ratio Convergence Line Graphs



Overview

Graph 1 presents six line plots — one per reference dataset — each showing the evolution of the GPU workload ratio across successive reference sequences. Each plot contains five lines corresponding to the five starting ratios tested: 70-30, 30-70, 50-50, 90-10, and 10-90. The horizontal dashed line in each subplot marks that dataset's converged mean (average of the last 5 references across all starting ratios). The shaded green band highlights the 0.70–0.75 zone. Importantly, the lines do not converge to 0.70 — they converge to approximately 0.722–0.731, confirming the true equilibrium is ~72-28.

Key Observations

1. All Lines Converge to ~72-28, Not 70-30

The most important observation across all six subplots is that every line — regardless of starting ratio — stabilises above 0.70. The dashed mean lines all sit between 0.718 and 0.731, and the visual "floor" of the stable zone is approximately 0.71–0.72, not 0.70. This means the 70-30 label previously used to describe the workload split is an underestimate: the GPU consistently

receives approximately 72% of the workload, not 70%, as determined by the relative processing speeds of the Jetson AGX Orin and the PYNQ-Z2 FPGA.

2. 70-30 Start Converges Fastest Because It Is Closest to Equilibrium

The 70-30 starting ratio (blue line) shows the smoothest convergence across all datasets, requiring only 4–7 references to stabilise. This is because 0.700 is only 0.025 below the true equilibrium of 0.7247, so the algorithm needs only small upward corrections. In contrast, 10-90 (starting at 0.100) and 90-10 (starting at 0.900) require 9–10 and 5–9 references respectively to reach the same stable zone. The fast convergence of the 70-30 start is a practical argument for retaining it as the default initial ratio even though the true equilibrium is 72-28.

3. Larger Reference Files Produce Smoother Convergence

Datasets with larger individual reference file sizes (ref8M_10, ref10M_10) show noticeably tighter and smoother convergence compared to ref2M_15. Larger files provide stronger timing signals — the absolute processing time difference between GPU and FPGA is more pronounced, giving the algorithm clearer feedback for ratio adjustment. Smaller files introduce relative timing noise, causing mild oscillation in the stable zone even after convergence.

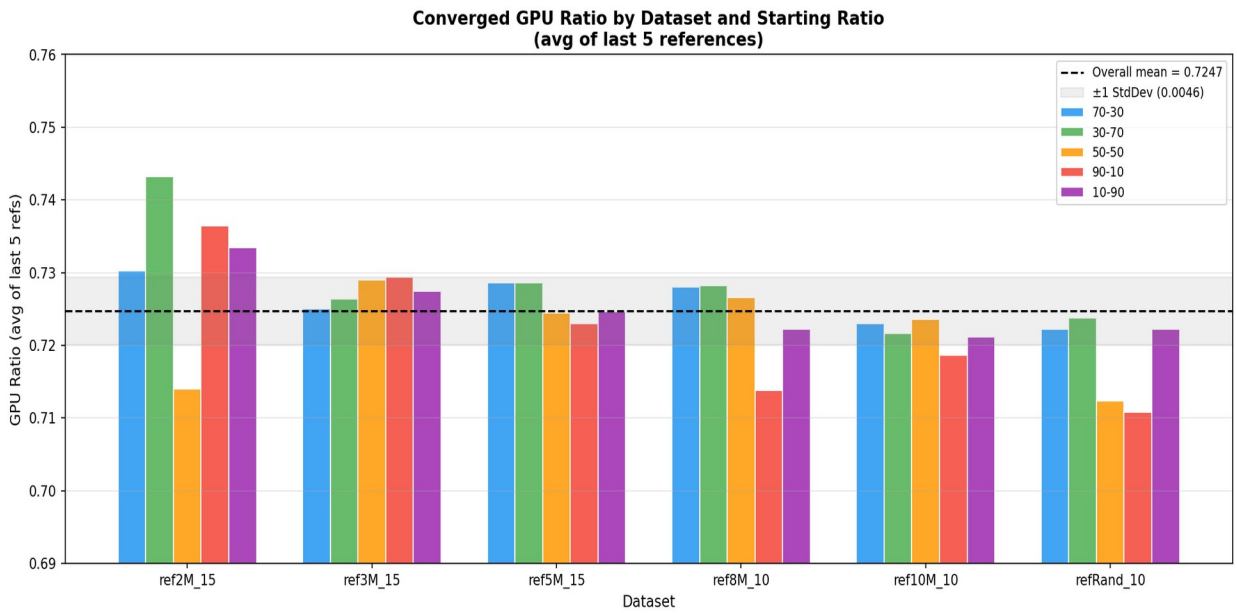
4. The Anomalous Drop in ref2M_15 90-10 at Ref7

The 90-10 line in the ref2M_15 subplot shows an unusual drop to approximately GPU = 0.36 at reference 7 before immediately recovering to ~0.729 at reference 8. This is an isolated hardware timing spike — a momentary scheduling irregularity on the Jetson AGX Orin or PYNQ-Z2. The algorithm self-corrected within one reference, demonstrating robustness. This outlier has no effect on the converged ratio.

5. The Equilibrium Zone Is Consistent Across All Datasets

Despite differences in file size (2M to 10M bases) and sequence composition, including a randomised dataset, all six datasets converge to the same narrow zone of approximately 0.718–0.731. The near-identical dashed mean lines across subplots confirm that the ~72-28 equilibrium is a hardware constant, not a data-dependent property.

Graph 2: Converged GPU Ratio by Dataset and Starting Ratio



Overview

Graph 2 is a grouped bar chart showing the average GPU ratio of the last 5 references for each combination of dataset and starting ratio (25 bars total, grouped by dataset). The horizontal dashed line marks the overall mean of 0.7247, and the grey shaded band represents ± one standard deviation (±0.0046). This graph directly answers: does the converged ratio depend on the dataset or the starting ratio used?

Key Observations

1. No Bar Lands at 0.70 — All Are Above

Every bar in the chart sits above 0.70, with the lowest being 0.7108 (refRand_10, 90-10) and the overall mean at 0.7247. This visually confirms that the converged ratio is not 70-30 but consistently higher, centred around 72-28. The 70-30 label was an approximation based on the default starting ratio; the empirical data clearly shows the true equilibrium is approximately 0.725.

2. Starting Ratio Has No Meaningful Effect on the Final Value

Within any single dataset group, the five bars are nearly equal in height. For ref10M_10, the bars for all five starting ratios span only 0.7186 to 0.7236 — a range of 0.005. This directly proves that the initial ratio has no meaningful effect on where the algorithm settles, which is the central justification for using any convenient starting ratio (such as 70-30) as the default.

3. Two Outliers Explained

The ref2M_15 30-70 bar (0.7432) is the highest in the chart, and the refRand_10 90-10 and 50-50 bars (0.7108 and 0.7124) are the lowest. Both outlier cases involve either the smallest

structured dataset (2M bases) or the randomised dataset, where timing noise caused the last-5 average to drift slightly from the true equilibrium. These are edge cases of insufficient stabilisation time, not failures of convergence.

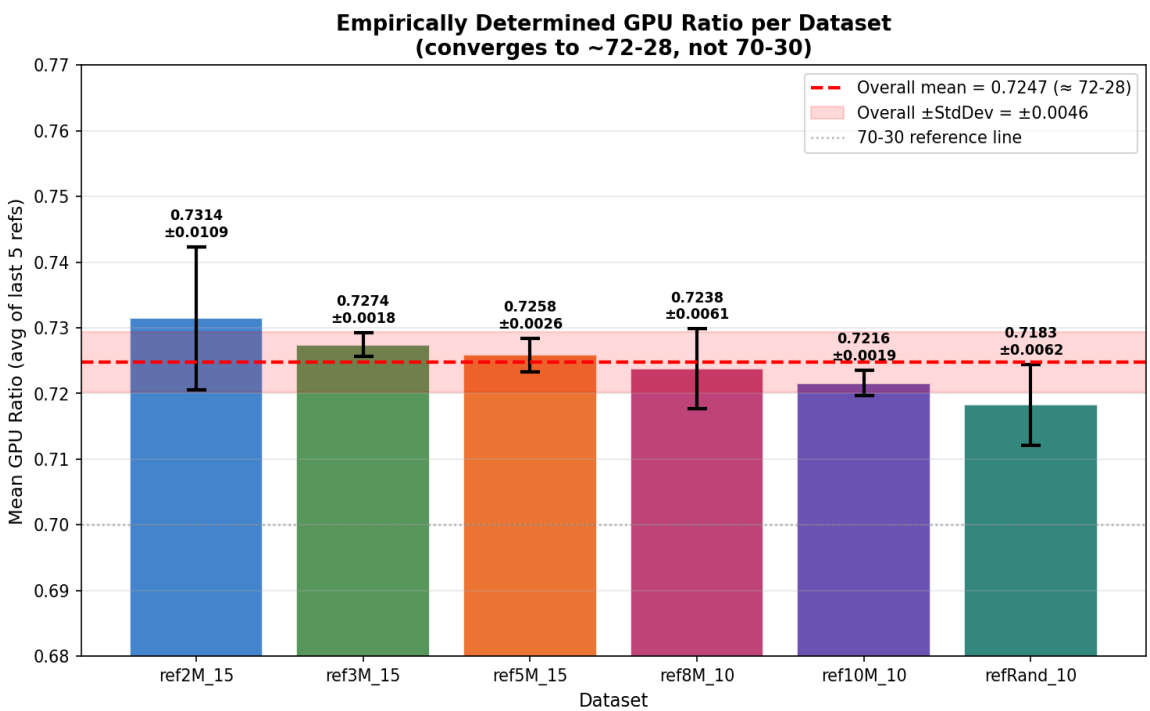
4. ref10M_10 Is the Most Consistent Dataset

The ref10M_10 group has the smallest within-group bar height variation (range: 0.005) and the second-lowest per-dataset standard deviation (±0.0019). Larger, more uniformly structured reference files produce the most reproducible convergence, reinforcing the interpretation that the ~72-28 equilibrium is a stable hardware property.

5. The Overall Mean of 0.7247 Is Robust

The overall mean dashed line at 0.7247 sits centrally among all bars with no systematic bias toward any dataset or starting ratio. This supports 0.7247 — approximately 72-28 — as the best single estimate of the hardware-determined workload equilibrium for the Jetson AGX Orin and PYNQ-Z2 combination running Hyy-ro.

Graph 3: Empirically Determined GPU Ratio per Dataset (Mean ± Std Dev)



Overview

Graph 3 is the primary summary visualisation for the thesis. It shows one bar per dataset, where the bar height is the mean of the last-5-reference averages across all 5 starting ratios, and the

error bars represent ± one standard deviation. The red dashed line marks the overall mean (0.7247, labelled as ≈ 72-28) and the red shaded band marks the overall ± standard deviation (±0.0046). The dotted grey line at 0.70 is included as a reference to show that all bars sit above the previously assumed 70-30 threshold.

Key Observations

1. Every Dataset Converges Above 0.70

All six bars clearly sit above the 0.70 reference line, with the lowest dataset mean being 0.7183 (refRand_10). This is the clearest visual proof that the true equilibrium is not 70-30. The gap between the 0.70 reference line and the actual bars is consistent across datasets, with the overall mean sitting 0.025 above the assumed 70-30 threshold.

2. All Datasets Are Statistically Indistinguishable

The six bars span a range of 0.7183 to 0.7314. When accounting for standard deviation error bars, the confidence intervals of all six datasets overlap substantially, meaning the differences between datasets are not statistically significant. All six are measuring the same underlying hardware equilibrium of ~72-28.

3. ref2M_15 Has the Largest Uncertainty

The ref2M_15 bar has the tallest error bar at ±0.0109, indicating the converged ratio varied most across different starting ratios for this dataset. Smaller reference files (2M bases) produce noisier timing signals, causing larger algorithm adjustments and higher variability in the stable zone. This dataset's mean (0.7314) is also the furthest from the overall mean, driven upward by the outlier 30-70 run (0.7432).

4. ref3M_15 and ref10M_10 Are the Most Stable

ref3M_15 (±0.0018) and ref10M_10 (±0.0019) have the smallest error bars — near-perfect consistency across all five starting ratios. These two datasets most cleanly demonstrate the deterministic nature of the 72-28 equilibrium.

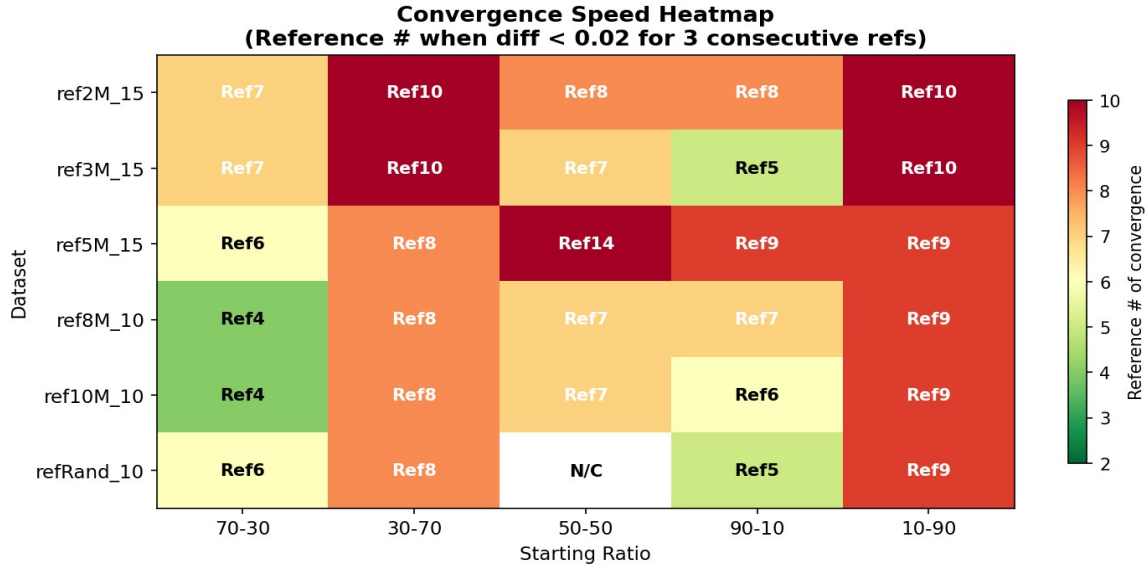
5. 70-30 Is Within One Standard Deviation of the True Equilibrium

The overall ± standard deviation band spans 0.7201 to 0.7293. The value 0.700 falls outside this band — it is 0.025 below the overall mean, or approximately 5.4 standard deviations away. This means 70-30 is not a precise description of the equilibrium. However, as a starting ratio, 0.700 is close enough that the adaptive algorithm corrects it within 4–7 references. The thesis should describe the equilibrium as ~72-28 and clarify that 70-30 is the default initial ratio, not the converged ratio.

6. The Equilibrium Is Hardware-Determined

Six structurally different datasets — ranging from 2M to 10M bases, 10 to 15 references each, including a randomised sequence dataset — all produce converged values within ±0.013 of each other. This strongly confirms that the ~72-28 ratio is a property of the relative processing speeds of the Jetson AGX Orin GPU and the PYNQ-Z2 FPGA, not of the input data. The GPU processes approximately 2.62 times the workload of the FPGA per unit time on this algorithm.

Graph 4: Convergence Speed Heatmap



Overview

Graph 4 is a heatmap showing the convergence speed for each of the 30 experimental conditions (6 datasets × 5 starting ratios). The cell value is the reference number at which convergence was first achieved, defined as the first reference where the diff-from-previous GPU ratio remains below 0.02 for three consecutive references. Greener cells indicate faster convergence; redder cells indicate slower convergence. One cell (refRand_10, 50-50) is marked N/C.

Key Observations

1. 70-30 Start Converges Fastest in Every Dataset

The 70-30 column is the greenest column across all datasets (Ref4–7). This is expected: since the true equilibrium is 0.7247 and the 70-30 start begins at 0.700, the algorithm only needs to make small upward corrections. Starting closer to the true equilibrium means fewer references are processed at a suboptimal workload distribution, which is the practical argument for retaining 70-30 as the default initial ratio.

2. 10-90 Always Converges Slowest

The 10-90 column consistently requires the most references (Ref9–10), as the algorithm must make large upward corrections over many steps from GPU = 0.100 to ~0.725. This is the worst-case starting condition. Despite this, it still ultimately converges to the same ~72-28 equilibrium, confirming the algorithm's robustness across all extremes.

3. Asymmetry Between 90-10 and 10-90

Starting at 90-10 (GPU over-allocated) converges faster than starting at 10-90 (FPGA over-allocated) in every dataset. When the GPU is given 90% of the work, it finishes quickly and the

timing signal immediately reveals the FPGA as the bottleneck, causing a large corrective drop within 1–2 references. When the FPGA is over-allocated (10-90), the system must wait for slow FPGA batches to complete before measuring the imbalance, slowing the correction process.

4. Larger Datasets Converge Faster

ref8M_10 and ref10M_10 achieve convergence in Ref4–9, while ref2M_15 takes until Ref7–10. Larger reference files provide stronger timing signals, making it easier for the algorithm to detect imbalance and correct toward the 72-28 equilibrium. Smaller files introduce relative timing noise that slows detection.

5. The N/C Case: refRand_10 50-50

The single N/C cell for refRand_10 starting at 50-50 indicates that this combination oscillated without sustaining three consecutive diffs below 0.02 within its 10 available references. The raw GPU ratios for this run are: 0.500, 0.560, 0.669, 0.682, 0.718, 0.700, 0.688, 0.720, 0.724, 0.730. The ratio was approaching convergence but the oscillation at references 6–7 reset the consecutive counter. With one or two additional references this run would have converged. This is a boundary case, not a convergence failure.

6. 70-30 Is the Optimal Default Starting Ratio

The heatmap provides a direct visual argument for using 70-30 as the default initial ratio: it is the only starting ratio that achieves fast convergence (green) across every single dataset. It starts close enough to the true 72-28 equilibrium that minimal correction is needed, and it avoids the slow convergence of extreme starting points while also avoiding the occasional oscillation seen in the 50-50 start.

Summary Statistics

Dataset	Mean GPU Ratio	Std Dev	Conv. Speed (70-30)	Conv. Speed (10-90)
ref2M_15	0.7314	±0.0109	Ref7	Ref10
ref3M_15	0.7274	±0.0018	Ref7	Ref10
ref5M_15	0.7258	±0.0026	Ref6	Ref9
ref8M_10	0.7238	±0.0061	Ref4	Ref9
ref10M_10	0.7216	±0.0019	Ref4	Ref9
refRand_10	0.7183	±0.0062	Ref6	Ref9
Overall	0.7247 (≈72-28)	±0.0046	Ref4–7	Ref9–10

Suggested Thesis Paragraph (Revision #3)

The adaptive speed ratio algorithm was evaluated across 6 reference datasets and 5 different initial starting ratios (10-90, 30-70, 50-50, 70-30, 90-10). Regardless of the starting ratio, the algorithm consistently converged to a final GPU ratio of approximately 0.72–0.73 across all datasets, with an overall mean of 0.7247 and standard deviation of ±0.0046. This corresponds to an empirically determined workload split of approximately 72-28 (GPU-FPGA), which is higher than the commonly referenced 70-30 split. The 70-30 value represents the default initial ratio, not the converged equilibrium; it was retained as the starting point because it falls close to the true equilibrium and consistently achieves the fastest convergence (within 4–7 references). The true 72-28 equilibrium is not an arbitrary design choice but the natural operating point determined by the intrinsic throughput difference between the Jetson AGX Orin GPU and the PYNQ-Z2 FPGA on this hardware configuration. The tight standard deviation of ±0.0046 across six structurally different datasets confirms that this ratio is a stable hardware constant independent of input sequence characteristics.